

Unit 01

Limitations of Bioplastics in Solving the Plastic Pollution Crisis

A. Warming Up

- What comes to mind when you think about the issue of plastic pollution?
바다에 떠다니는 플라스틱, 쓰레기장, 자연을 파괴하는 오염
- Have you ever tried to reduce your own plastic consumption? If so, what steps did you take?
식료품통 대신 공짜 장수의 텀러리용 사물함
플라스틱 병 대신 종이 병 사용
- What are bioplastics, and why are they considered an important solution to the plastic pollution problem?
생분해성 플라스틱, 자연적으로 분해되는 플라스틱
생분해성, 친환경, 지속 가능한 소재

B. Key Expressions

- Despite** efforts from industry, governments, and NGOs, the plastic pollution crisis is worsening.

Assignment 01

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D. Comprehension Checkup

Choose the best answer to the following questions.

- What is the main challenge in creating affordable and effective bioplastics?
(A) Lack of demand for bioplastics
(B) Limited availability of raw materials
(C) Difficulty in achieving complete decomposition
(D) High cost of production
- What is one of the strategies proposed in the Pew report to address plastic waste?
(A) Minimizing the use of plastic packaging
(B) Banning the export of waste plastic
(C) Increasing plastic consumption
(D) Encouraging the use of single-use plastics
- Why is extended producer responsibility considered important in addressing plastic waste?
(A) It shifts the financial burden of recycling onto governments.
(B) It ensures that consumers take responsibility for recycling.
(C) It promotes reusing and recycling plastic products.
(D) It limits the use of traditional plastic packaging.

Science & Technology Reading

- Why do bioplastics like PHA and PLA struggle to match the properties of traditional plastic?
(A) They are made from expensive raw materials.
(B) They are difficult to decompose completely.
(C) They lack strength and versatility.
(D) They are not cost-effective to produce.
- What is the primary source of plastic entering the oceans?
(A) Industrial waste
(B) Landfills
(C) Rivers in Asia and Africa
(D) Plastic manufacturing facilities
- What is necessary for effective recycling of plastic bottles?
(A) Dependence on governments to establish recycling infrastructure
(B) Adoption of circular economy principles by companies
(C) Promotion of bioplastics as a replacement for traditional plastics
(D) Increased plastic consumption to meet demand

E. Grammar Checkup

Choose one that can best complete the following sentences.

- The idea of a plastic that can be used, discarded anywhere, and safely decompose completely _____ not currently exist, even in nature.
(A) does
(B) do
(C) is
(D) are

- The key to _____ plastic waste lies in a comprehensive transformation of the global economy towards recycling much larger quantities of plastic than currently practiced.
(A) address
(B) addressing
(C) be addressed
(D) being addressed
- Around 11 million metric tons of plastic _____ the oceans annually, three million more than previous estimates.
(A) enter
(B) are entered
(C) to enter
(D) to be entered
- Shifting the financial burden of recycling onto the producers _____ gaining momentum, with companies like Nestle Waters committing to support extended producer responsibility in the beverage industry.
(A) be
(B) to be
(C) is
(D) are
- The Great Pacific Garbage Patch has grown to four times the size of California, _____ a severe threat to ecosystems.
(A) pose
(B) to pose
(C) posing
(D) posed
- Skeptics argue that more specific details about these developments need to be disclosed for _____ to be considered viable solutions.
(A) it
(B) you
(C) that
(D) them

F. Vocabulary Checkup

Choose one that can best complete the following sentences.

- Bioplastics, including Coke's PlantBottle, have been _____ as a potential solution to the world's plastic pollution problem.
(A) imported (B) recycled
(C) rejected (D) hailed
- _____ efforts from industry, governments, and NGOs, the plastic pollution crisis is worsening.
(A) despite (B) due to
(C) for the purpose of (D) in terms of
- Creating _____ and effective bioplastics has proven to be a significant challenge.
(A) sustainable (B) efficient
(C) affordable (D) variable
- The report outlines various strategies, including minimizing the use of plastic packaging by _____ it with paper or compostable materials.
(A) excluding (B) including
(C) substituting (D) duplicating
- Bioplastics, once thought to be a promising solution, have _____ expectations.

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- (X) fallen short of (B) paid attention to
(C) taken care of (D) contributed to

- To enable effective recycling, companies like Coca-Cola and Pepsi must commit to selling beverages only in bottles containing recycled content, _____ the cost.
(X) regardless of (B) in addition to
(C) with respect to (D) because of

G. Guided Composition

Using the useful expressions given, put the following Korean into English.

- 바이오 플라스틱은 전 세계 플라스틱 오염 문제를 해결할 수 있는 잠재적 해결책으로 주목받고 있다.
(a potential solution to)
bioplastics have been hailed as a potential solution
to the world's plastic pollution problem
- 저렴하고 효과적인 바이오 플라스틱을 만드는 것은 상당히 어렵다는 것이 입증되었다.
(have proven to be)
creating affordable and effective bioplastics has
proven to be a significant challenge.

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- 플라스틱 폐기물의 양은 2040년까지 3배로 증가할 것이다. (triple)
The amount of plastic waste will triple by 2040.

- 포장재를 사용하는 기업은 제품을 재활용하고 재사용하는 데 앞장서야 한다. (take the lead)
companies that use packaging materials must take
the lead in recycling and reusing their products.

- 한때 유망한 해결책으로 여겨졌던 바이오 플라스틱은 기대에 미치지 못했다. (fall short of)
bioplastics are thought to be a promising solution,
have fallen short of expectations.

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